

Modeling Citrus Irrigation Requirements Using Environmental Drivers

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```
#| label: setup #| include: false

import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns
import sys, os from pathlib import Path from IPython.display import Markdown

sys.path.insert(0, os.path.join(os.getcwd(), ".."))

from src.generate_data import generate_citrus_data from src.viz import ( plot_correlation_heatmap,
plot_temp_vs_water, plot_growth_stage_box, plot_feature_importance, )

from sklearn.model_selection import train_test_split, cross_val_score from sklearn.preprocessing
import LabelEncoder from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
from xgboost import XGBRegressor

data_path = Path("../data/processed/citrus_water.parquet")

if data_path.exists(): df = pd.read_parquet(data_path) else: df = generate_citrus_data()

np.random.seed(42)

““markdown # Introduction
```

0.1 Problem Statement

Efficient irrigation management is critical for citrus production under increasing climate variability. Over-irrigation wastes resources and increases disease risk, while under-irrigation reduces yield and fruit quality.

This study aims to:

- Quantify drivers of daily water demand (L/tree/day)
- Build a predictive regression model
- Provide operational irrigation guidance

0.2 Dataset Description

The dataset contains 2,000 daily observations including:

- Temperature (°C)
- Humidity (%)
- Wind speed (km/h)

- Rainfall (mm)
- Tree age (years)
- Soil type (categorical)
- Growth stage (categorical)
- Target: water_need_liters

```
Markdown(df.describe().round(2).to_markdown())
```

```
““markdown ## Correlation Structure
```

```
““tetibusq #| label: fig-corr #| fig-cap: “Pearson correlation matrix.”
```

```
fig_corr = plot_correlation_heatmap(df) plt.show()
```

```
““markdown Observation: Temperature exhibits the strongest linear association with water demand, followed by wind speed and humidity.
```

1 Exploratory Data Analysis

1.1 Temperature–Demand Relationship

“tetibusq #| label: fig-temp #| fig-cap: “Non-linear relationship between temperature and irrigation demand.”

```
fig_temp = plot_temp_vs_water(df) plt.show()
```

“markdown The relationship is non-linear and amplified under low humidity conditions.

1.2 Agronomic Segmentation

“tetibusq #| label: fig-growth #| fig-cap: “Distribution of water demand across growth stages and soil types.”

```
fig_growth = plot_growth_stage_box(df) plt.show()
```

“markdown **Key finding:** Fruiting-stage trees on sandy soils demonstrate the highest median demand.

1.3 Lookup Aggregation

“tetibusq #| label: tbl-lookup #| tbl-cap: “Median water demand by temperature band and growth stage.”

```
bins = pd.cut(df[“temperature_c”], bins=[0, 20, 30, 40, 50], labels=[“<20°C”, “20-30°C”, “30-40°C”, “>40°C”])
```

```
lookup = ( df.assign(temp_band=bins) .groupby([“growth_stage”, “temp_band”], observed=True) [“water_need_liters”] .median() .round(1) .unstack() )
```

```
Markdown(lookup.to_markdown())
```

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2 Modeling Methodology

2.1 Target Variable Formulation

To ensure agronomic validity, the target variable ($y = \text{Water Need}$) was synthesized based on the **FAO-56 Penman-Monteith framework**, the global standard for irrigation scheduling.

Due to the practical constraints of sensor availability (specifically solar radiation), **Reference Evapotranspiration** (ET_0) was estimated using the **Hargreaves-Samani equation**, a temperature-based approximation recommended by ASCE and FAO for data-scarce regions:

$$ET_0 = 0.0023 \cdot (T_{mean} + 17.8) \cdot \sqrt{T_{max} - T_{min}} \cdot R_a$$

The final Crop Evapotranspiration (ET_c) was derived as:

$$ET_c = ET_0 \times K_c \times K_s - P_{eff}$$

Where:

- K_c (**Crop Coefficient**): Adjusted for citrus phenological stages:
 - *Vegetative*: 0.65
 - *Flowering*: 0.85
 - *Fruiting*: 1.00 (Peak demand)
- K_s (**Soil Stress Coefficient**): Modifies uptake based on soil retention (Sandy: 1.2, Clay: 0.85).
- P_{eff} (**Effective Rainfall**): Accounted for as 80% of daily precipitation.

Methodological Note

While the full Penman-Monteith equation is the theoretical ideal, the Hargreaves-Samani method correlates ($r > 0.95$) effectively in semi-arid climates and provides a robust ground-truth for supervised learning when radiometric data is unavailable.

2.2 Feature Engineering

Categorical variables were label-encoded.

```
“tetibusq #| label: feature-engineering
```

```
from sklearn.preprocessing import LabelEncoder from sklearn.model_selection import train_test_split
```

```
df_model = df.copy()
```

```
le_soil = LabelEncoder() le_stage = LabelEncoder()
```

```
df_model[“soil_type_enc”] = le_soil.fit_transform(df_model[“soil_type”]) df_model[“growth_stage_enc”]  
= le_stage.fit_transform(df_model[“growth_stage”])
```

```
features = [“temperature_c”, “humidity_pct”, “wind_speed_kmh”, “rainfall_mm”, “tree_age_years”,  
“soil_type_enc”, “growth_stage_enc”,]
```

```
X = df_model[features] y = df_model[“water_need_liters”]
```

```

X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42 )
print(f"Training samples: {len(X_train)}") print(f"Testing samples: {len(X_test)}")
pythontetibusq #| label: tbl-features #| tbl-cap: "Model Input Features" #| echo: false
feature_metadata = pd.DataFrame({ "Feature Name": [ "Temperature", "Humidity", "Wind
Speed", "Rainfall", "Tree Age", "Soil Type", "Growth Stage" ], "Data Type": [ "Float", "Float",
"Float", "Float", "Integer", "Categorical (Encoded)", "Categorical (Encoded)" ], "Unit / De-
scription": [ "Daily Mean (°C)", "Relative (%)", "km/h", "mm/day", "Years since planting",
"Sandy/Loam/Clay", "Phenological stage" ] })

Markdown(feature_metadata.to_markdown(index=False))

"markdown ## Model Selection

We selected XGBoost Regressor due to:



- Non-linear modeling capacity
- Robustness to multicollinearity
- High predictive performance



"tetibusq #| label: model-training

model = XGBRegressor( n_estimators=300, max_depth=5, learning_rate=0.08, random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

r2 = r2_score(y_test, y_pred) mae = mean_absolute_error(y_test, y_pred) rmse =
np.sqrt(mean_squared_error(y_test, y_pred))

cv_scores = cross_val_score(model, X, y, cv=5, scoring="r2")

print("Test R²:", round(r2, 3)) print("MAE (L):", round(mae, 2)) print("RMSE (L):", round(rmse,
2)) print("Cross-validated R²:", round(cv_scores.mean(), 3))

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```

3 Model Interpretation & Recommendations

3.1 Feature Importance

“tetibusq #| label: fig-importance #| fig-cap: “Normalized feature importance from XGBoost.”

```
importance = dict(zip(features, model.feature_importances_)) fig_imp = plot_feature_importance(importance)
plt.show()
```

“markdown Temperature is the dominant explanatory variable, followed by humidity and wind speed.

3.2 Operational Implications

Based on model outputs:

1. Irrigation demand increases exponentially beyond 30°C.
2. Fruiting-stage trees require ~30–50% more water than vegetative stage.
3. Sandy soils amplify irrigation sensitivity.
4. Wind >15 km/h increases evapotranspiration effects.

3.3 Limitations

- Synthetic dataset (not field sensor data)
- No seasonal time-series modeling
- No soil moisture sensor integration

3.4 Future Work

- Incorporate real ET meteorological data
- Add time-series forecasting
- Deploy real-time dashboard (Marimo or Streamlit)
- Integrate IoT soil moisture measurements

4 Conclusion

This whitepaper demonstrates that environmental and agronomic variables can predict citrus irrigation requirements with high accuracy ($R^2 > 0.9$). The framework enables scalable, data-driven irrigation scheduling and can be extended to additional crops and climatic regions.